Ruitong Tian

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EDUCATION

- **University of Florida** — *2025 – Present*, Gainesville, FL
  Ph.D. student | Sustainable Construction. Advisor: Eric Jing Du.
  Research focus: Dexterous manipulation, robot learning, and tactile sensing.
- **National University of Singapore** — *2023 – 2025*, Singapore
  M.S. & Year 4 Undergraduate Exchange | Infrastructure Digitalization & Management.
  Advisor: Prof. Poh Leong Hien.
- **South China University of Technology** — *2020 – 2024*, Guangzhou, China
  B.E. | Green Intelligent Construction. Advisor: Prof. Nan Hu.

PROJECTS & PUBLICATIONS

ExoBridge: Learning a Bare Hand to Hand-Worn Exoskeleton Mapping through Human Limb Coupling — **Under submission**, *University of Florida*
Ruitong Tian, Xianyao Li, Noah B. Wilson, Fang Xu, Eric Jing Du.
- Developed ExoBridge to map bare-hand RGB video to exoskeleton motion, contact, and tactile intensity.
- Evaluated 1,215 demonstrations from 31 sessions across four tasks: contact AUROC 0.916, intensity correlation 0.790, and 44.7% lower motion error (primary channels) versus a zero-change reference.

From Wearable Interfaces to Dexterous Policies: Contact Shifts and Tactile Representations — **Under submission**, *University of Florida*
Ruitong Tian, Fang Xu, Noah B. Wilson, Xianyao Li, Eric Jing Du.
- Studied wearable design and tactile inputs for Diffusion Policy on XHand across four manipulation tasks.
- Tested 240 rollouts: spatial contact/force yielded 52% pooled success versus 35% for binary contact.

Semantically Enhanced Eye-Tracking Controls in Human-Robot Interaction: A Comparison Study — **Ergonomics**, **Accepted Oct. 9, 2026**, *University of Florida*
Ruitong Tian, Hengxu You, Fang Xu, Jiahao Wu, Bowen Sun, Thanakon Uthai, Eric Jing Du.
- Co-designed a 30-participant study of gaze, voice, and gesture control with HoloLens 2 and Franka Emika.
- Measured mean per-step times of 17.10 s (gaze) versus 32.78 s (hand gesture); analyzed quality-screened gaze data from 19 participants and 96 trials.

Nudge Before You Push: Physics-Aware Navigation via Tactile Probing — **Under submission**, *University of Florida*
Xianyao Li, Fang Xu, **Ruitong Tian**, Bowen Sun, Xiao Hu, Yang Ye, Jing Du.
- Coauthored TANav; simulation missed-push rate was 5.5% versus 28.7% at comparable low-risk operating points.

Controlling crack propagation in layered beams with architected lattice-reinforced composite interlayer designs — **Construction and Building Materials (2024)**, *South China University of Technology*
Binglin Xie, **Ruitong Tian**, Haiyang Zhao, Tongtong Ye, Yuan Zhang, Nan Hu.
- Designed and 3D-printed lattices; validated Abaqus crack models with fracture/fatigue tests and DIC.

ADDITIONAL EXPERIENCE & AWARDS

Machine Learning for Thermal-Image Damage Prediction — *National University of Singapore*
- Proposed a façade crack detection framework; built PyTorch CNN/knowledge-transfer models for damage severity prediction (correlation 0.98); automated 2,000+ Abaqus samples and processed 150+ thermal images.
- **Leadership:** Main Initiator & Vice President, Intelligent Digital Engineering Association (2022–2023); debugged 3D printers and led printing/Cura training.
- **Awards:** Outstanding Undergraduate Thesis (Top 10%); Third Prize, Asia and Pacific Mathematical Contest in Modeling (2023).

TECHNICAL SKILLS

- **Programming:** Python, C++, MATLAB.
- **Machine Learning:** PyTorch, Hugging Face, scikit-learn, imitation learning, Diffusion Policy, CNNs, temporal modeling.
- **Computer Vision & Data:** OpenCV, DINOv2, FastSAM, MediaPipe, NumPy, pandas, SciPy.
- **Robotics & Simulation:** ROS/ROS 2, MuJoCo, Isaac Sim, dexterous manipulation, tactile sensing.
- **Robotic Hardware & Platforms:** DexUMI, XHand, Franka Emika, HoloLens 2.
- **Systems & Computing:** Linux (Ubuntu), Git, CUDA, Slurm, HiPerGator, multi-node GPU training.